HP Spectre XT was one of the better ultrabooks we tested this year

the improvements, some tradeoffs are present in the ultrabook such as lack of an optical drive, the audio output from the built-in speakers not being good enough and some of them come with plasticky build to keep the weight light.

With the Windows 8 fever slowly gripping everyone, there are interesting form factors such as hybrids, slates, convertibles and so on. It will be interesting to see how ultrabooks compete against these new form factors in 2013.

Overhauled operating systems (Good)

The integration of content between the smartphone and the desktop computer was inevitable, thanks to the powerful innards of the smartphones these days. Phones today have more compute power than PC of just a few years ago.

This year can be called the year where we have seen the integration between the desktop OS and the mobile OS.

iOS and Mountain Lion integration pairs devices quite well. You can wirelessly sync content from your iOS device to your Mac, be it music, videos, photos, iMessages and even notes, all simply via Wi-Fi. Click a picture from your iPhone and watch it pop up on your Mac in a matter of seconds. The integration of iOS and Mountain Lion even lets users play multiplayer games with friends on a Mac, iPhone, iPad, or iPod touch – across devices.

Windows 8 has seen the biggest rehash to an OS in Microsoft's history. Windows 8 too will see the integration of content between Windows Phone 8 and Windows 8, but as of now, the only integration we have seen is through Sky Drive which gives mobile users access to their documents and data stored on Sky Drive. Microsoft has also integrated users Xbox Live accounts throughout their devices. Users can earn achievement points by playing games on their portable devices and also have access to their Live avataars, modify them and go through their Xbox content on any of their devices.

Windows 8 is also bridging the gap between touchscreen devices and the laptop and we have started seeing devices emerge that offer the best of both worlds.

This technology is still in its nascent stages as this is the first year where we have seen a deep integration between the mobile and the desktop OS and it seems to be moving in the right direction.

The Mobile OS space too has seen some action. iOS 6 still has the same simple interface that people are accustomed to and that isn't a bad thing. Android on the other hand has seen some really awesome improvements. Android 4.1 Jelly Bean brought with it a slew of impressive features such as Chrome becoming the native browser, Android Beam, face unlock, better notifications, Project Butter making devices a lot more smoother and Google Now which gave users relevant information such as weather, traffic and more. With Android 4.2, devices have the ability to have multiple users on one device, wireless viewing of content on your TV (just like AirPlay), gesture typing on the keyboard and a whole lot more.

Android Jelly Bean launched a boatload of features this year

It's safe to say, that the mobile OS has seen some pretty big improvements this past 12 months. At this rate, the next year should be quite interesting in this space.

Integrated and discrete graphics (Good)

The traditional rivalry between the two giants from the graphics industry carried on this year as well. The GPU manufacturing process was reduced to 28nm by both the players. AMD rolled out the Radeon HD 7000 series towards the end of December 2011 and it was a good three-month wait before NVIDIA released its GTX 600 series of cards. But the wait was well worth it, as NVIDIA's flaghip card – GTX 680 – beat AMD's flagship HD 7970 in the performance arena, although by a small margin. But what impressed everyone was the power efficiency and features offered. The GTX 680 was quickly followed by GTX 670 which was again a better performer than AMD's HD 7950 and so on it continued. In fact, NVIDIA even released its dual-GPU beast –GTX 690 – which retailed at the price point of a good multimedia rig. We didn't get a chance to try out AMD's dual-GPU card, as it wasn't available here till the time of finalising our Zero1 awards. In the second half of the year, came the more mainstream cards with comparitively lower prices from both AMD and NVIDIA, where again NVIDIA kept winning accolades.

AMD introduced a new architechture with the HD 7000 series cards called Graphics Core Next (GCN) which did away with the Very Long Instruction Word (VLIW) architecture to make the GPUs based on the GCN architecture more compute-friendly. VLIW architecture is best for massive parallel processing tasks, but a compute-related task is different. Do read out February 2012 issue for an in-depth look at GCN.

NVIDIA introduced the Kepler architecture which introduced a new stream multiprocessor called SMX, each of which had 192 CUDA cores, as compared to 32 seen on the Fermi cards. Thanks to the 28nm manufacturing process, NVIDIA managed to keep the maximum TDP under 200 Watts, thereby making the graphics cards more power efficient.